

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.2	10.4	0.2	0	0	62.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.7	9	0.1	0	0	62.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.6	9.6	0.3	0	0	62.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.5	8.6	0.1	0	0	62.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.3	10.1	0.2	0.1	0	62.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	29	8.6	0.1	0	0	62.3
U2 - Efficacy additive alpha prior Beta(1, 1)	26.7	10.5	0.3	0	0	62.5
U3 - Efficacy additive alpha prior Beta(1, 1)	29.8	7.7	0.1	0	0	62.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	3.7	1.3	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.3	3.7	1.4	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.3	3.7	1.3	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.4	3.7	1.3	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.3	3.7	1.3	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.4	3.6	1.3	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	16.4	3.7	1.3	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	16.4	3.6	1.3	0.3	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	1.9	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	1.9	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	1.9	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	1.9	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	1.9	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	1.9	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	1.9	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	1.9	0.4	0.2	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	36	22.8	30.5	7.5	1.3	1.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	39.5	23.6	26.7	7.5	0.8	1.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	36.5	22.2	30.7	7.3	1.4	1.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	39.9	22.3	28	7.1	0.8	1.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	38	19.1	31.5	8.2	1.3	1.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	41.9	19.1	29.1	7	1	1.9
U2 - Efficacy additive alpha prior Beta(1, 1)	38.2	19	31.5	8.3	1.1	1.9
U3 - Efficacy additive alpha prior Beta(1, 1)	41.5	20.7	28.4	6.4	1	2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.5	6	6.9	3.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.5	6.1	6.7	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.6	5.9	6.8	3.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.6	5.9	6.8	3.2	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.6	5.8	7	3.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.7	5.8	7	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.7	5.8	6.9	3.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.7	5.8	6.8	3.3	1.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.6	1.2	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.6	1.2	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.6	1.2	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.6	1.2	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.5	1.3	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.5	1.3	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.6	1.2	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.6	1.2	0.7	0.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.7	46.3	17.9	6.2	0.5	10.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.7	48.4	16.2	5.1	0.3	10.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.8	46.1	19.2	5.9	0.6	10.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.9	46.6	17.8	4.9	0.4	10.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.1	44	20	5.9	0.7	10.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21	45.3	18.1	4.8	0.3	10.5
U2 - Efficacy additive alpha prior Beta(1, 1)	18.7	44.2	19.9	6.3	0.5	10.4
U3 - Efficacy additive alpha prior Beta(1, 1)	20.9	44.7	18.4	5.5	0.2	10.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	8.3	5.6	3	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	8.4	5.6	3	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	8.2	5.8	3	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	8.1	5.8	3	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	8	5.8	2.9	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	8.1	5.8	3	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.8	8	5.8	2.9	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.9	8	5.8	2.9	1.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.1	1.1	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.1	1.1	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3	1.1	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3	1.1	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3	1.1	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3	1.1	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3	1.1	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3	1.1	0.8	0.2	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	9.5	23.9	44	11.9	1.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.7	11.5	20.3	42.3	10.3	1.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9	10.2	23.5	43.1	12.3	1.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14	11.3	20.7	42.3	9.8	1.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.1	9.8	24.3	43	12.9	1.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.9	9.7	21.4	42	11.1	1.9
U2 - Efficacy additive alpha prior Beta(1, 1)	7.4	9.7	25.5	43.1	12.4	1.9
U3 - Efficacy additive alpha prior Beta(1, 1)	14.4	10	21	41.4	11.3	1.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.8	4.7	7.1	8.1	4.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.8	4.8	7	8.2	4.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	4.7	7.1	8.2	4.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	4.7	7.1	8.3	4.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	4.7	7.1	8.2	5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	4.7	7.1	8.2	5	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.8	4.7	7.1	8.2	4.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.8	4.6	7.1	8.2	5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	1.5	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	1.5	2.3	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	1.5	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3	1.5	2.3	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3	1.5	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3	1.5	2.2	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	1.5	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	1.5	2.2	0.7	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	35.6	26.3	21.5	8	4.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	42.3	23.4	16.7	5.6	4.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	35	26.8	20.9	8.6	4.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8	40.6	24.7	17	5.5	4.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	32	29.2	20	9.1	4.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	37.4	28.6	15.8	5.3	4.2
U2 - Efficacy additive alpha prior Beta(1, 1)	5.4	31.9	29.4	20.5	8.6	4.2
U3 - Efficacy additive alpha prior Beta(1, 1)	8.9	37.7	27.9	15.5	5.8	4.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	7	6.7	6	4.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	7	6.7	6	4.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.2	6.8	6.8	6	4.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.2	6.8	6.7	6	4.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.2	6.7	6.9	5.9	4.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.2	6.8	6.8	5.9	4.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.2	6.7	6.8	5.9	4.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.2	6.8	6.8	5.9	4.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.4	1.7	2.2	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.4	1.7	2.1	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.4	1.6	2.2	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.4	1.6	2.1	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.6	2.1	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.6	2.1	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.3	1.7	2.1	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.3	1.6	2.1	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.2	12.9	2.6	0.6	0.1	14.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	71.7	11.4	1.8	0.3	0.1	14.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	70.9	11.5	2.4	0.5	0.1	14.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	73	10	1.6	0.5	0.1	14.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	71.5	10.5	2.6	0.6	0.1	14.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	73.1	9.7	2.1	0.4	0.1	14.6
U2 - Efficacy additive alpha prior Beta(1, 1)	71.9	10.2	2.7	0.6	0.1	14.5
U3 - Efficacy additive alpha prior Beta(1, 1)	73.5	9.5	2.1	0.3	0.1	14.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.9	4	1.7	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.9	4	1.7	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21	3.9	1.8	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21	3.9	1.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.1	3.8	1.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21	3.8	1.8	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	21.1	3.8	1.8	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	21.1	3.8	1.8	0.6	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.5	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.5	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.5	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.5	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.5	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.5	0.4	0.2	0	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.3	44.6	17	6.8	1.9	2.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.7	46.2	14.9	6.7	1.1	2.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.4	42.3	19	7.4	1.5	2.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.3	45	15.6	6	0.7	2.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.4	40.3	18.6	8.7	1.6	2.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.6	42.9	15.8	7.6	0.7	2.4
U2 - Efficacy additive alpha prior Beta(1, 1)	28.3	40.9	18.5	8.3	1.6	2.4
U3 - Efficacy additive alpha prior Beta(1, 1)	31	43.4	15.6	6.8	0.8	2.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	7.8	5.2	3.2	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	7.8	5.3	3.2	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.9	7.7	5.4	3.1	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.9	7.7	5.3	3.2	1.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	7.5	5.4	3.2	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	7.6	5.4	3.2	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.9	7.5	5.4	3.2	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.9	7.6	5.4	3.2	1.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1	0.8	0.3	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	42.3	8.4	1.2	0.3	0	47.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	43.3	7.8	0.7	0.2	0	48
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	42.8	8.5	0.9	0.3	0	47.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	43.3	8	0.8	0.2	0	47.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	41.9	8.8	0.9	0.2	0.1	48.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	42.8	8.4	1	0.2	0	47.6
U2 - Efficacy additive alpha prior Beta(1, 1)	42.4	8.3	1.1	0.2	0	48
U3 - Efficacy additive alpha prior Beta(1, 1)	43.1	7.8	1	0.2	0	47.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.8	3.1	1.1	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.8	3.1	1.1	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.9	3.1	1.1	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.9	3.1	1.1	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.9	3	1.1	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.9	3.1	1.1	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.9	3	1.1	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	19	3	1.1	0.3	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.5	0.3	0.1	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.6	17.7	16.6	11.8	15.5	19.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.4	16	15.2	10.3	13.2	19.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.8	16.9	17.4	11.7	15.6	19.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.9	17	15	9.8	13.5	19.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.8	16.8	18.7	11.1	15.9	19.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25	16.8	15	9.9	13.7	19.6
U2 - Efficacy additive alpha prior Beta(1, 1)	17.4	17.3	18.2	12.1	15.3	19.7
U3 - Efficacy additive alpha prior Beta(1, 1)	26.7	15.6	14.6	9.8	13.6	19.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	5.1	5.5	3.7	3.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	5	5.4	3.7	3.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.9	5	5.5	3.7	3.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.9	5	5.4	3.7	3.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.9	5	5.4	3.6	3.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	4.9	5.4	3.7	3.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.9	4.9	5.5	3.7	3.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	10	5	5.5	3.6	3.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.8	1.3	1.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.8	1.3	1.3	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1.3	1.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.8	1.3	1.2	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.9	1.3	1.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.8	1.3	1.2	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	2.8	1.3	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	2.8	1.3	1.2	0.5	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13	33.4	8.4	2.4	0.8	42
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.6	34.9	7.3	2	0.8	41.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.3	32.3	9	2.4	0.8	42.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.7	33	8.4	1.9	0.8	42.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.2	32.7	9.9	2.4	1	41.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.6	32.1	8.3	2.2	0.8	42
U2 - Efficacy additive alpha prior Beta(1, 1)	12.7	32.6	9.4	2.5	1.1	41.7
U3 - Efficacy additive alpha prior Beta(1, 1)	13.7	33.2	8.3	2.1	0.9	41.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.2	6	3.3	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.2	6	3.3	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	5.9	3.4	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	6	3.3	1.5	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.1	5.9	3.5	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.1	5.9	3.4	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	13.2	6	3.4	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	13.2	6	3.4	1.5	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.3	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.3	0.7	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.3	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.3	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.3	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.3	0.7	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.3	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.3	0.7	0.4	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	66.4	24	2.1	0.1	0	7.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	68.8	23.1	0.8	0.1	0	7.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	67.9	23.1	1.7	0.1	0	7.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	69.9	21.6	1	0.1	0	7.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	69	21.8	1.6	0.1	0	7.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	72.2	19.8	0.4	0.1	0	7.5
U2 - Efficacy additive alpha prior Beta(1, 1)	69.7	21.5	1.4	0.1	0	7.3
U3 - Efficacy additive alpha prior Beta(1, 1)	73.3	18.5	0.8	0.1	0	7.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20	5.8	2.5	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20	5.9	2.4	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.2	5.7	2.4	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.1	5.7	2.5	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.2	5.5	2.5	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.1	5.6	2.5	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	20.2	5.6	2.5	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.3	5.5	2.5	0.5	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.3	0.6	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.4	0.6	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.4	0.6	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.3	0.6	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.3	0.6	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.3	0.6	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.3	0.6	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.3	0.6	0.2	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.2	15.5	12.4	36.6	16.1	3.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23	16.3	8.9	34	14.6	3.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.2	16.6	11.7	37.9	16.4	3.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.4	14.7	8.2	35.2	15.3	3.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.6	16	11.8	37.6	16.8	3.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.9	15.6	7.4	35.5	15.4	3.2
U2 - Efficacy additive alpha prior Beta(1, 1)	14.4	16.3	12.2	37.4	16.5	3.2
U3 - Efficacy additive alpha prior Beta(1, 1)	23.1	14.6	8.6	35.4	15.1	3.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.3	5.4	5.9	7.4	5.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.3	5.3	5.9	7.5	5.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.4	5.3	5.9	7.4	5.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	5.4	5.8	7.4	5.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	5.3	5.9	7.4	5.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.4	5.3	5.8	7.5	5.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.4	5.3	5.9	7.5	5.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.4	5.3	5.9	7.4	5.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.1	1.4	2.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.1	1.4	2.3	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.1	1.4	2.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.1	1.4	2.2	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.1	1.5	2.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.2	1.4	2.2	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.1	1.4	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.1	1.5	2.2	0.7	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.2	35	10.3	2.6	0.6	37.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.5	35.7	8.7	2.5	0.4	37.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.7	34.6	10.5	2.4	0.7	37.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.8	35.2	9	2.1	0.4	37.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.9	34.1	11.3	2.9	0.5	37.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.3	34.6	8.8	2.4	0.4	37.5
U2 - Efficacy additive alpha prior Beta(1, 1)	14.8	33.9	11.3	2.4	0.5	37.1
U3 - Efficacy additive alpha prior Beta(1, 1)	16.4	34.2	9.1	2.5	0.4	37.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.8	6.5	3.6	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.9	6.4	3.7	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.9	6.2	3.8	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.9	6.3	3.8	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13	6.1	3.9	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.9	6.3	3.8	1.5	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.9	6.2	3.8	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.9	6.2	3.8	1.5	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	0.8	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	0.8	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	0.8	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	0.8	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	0.8	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	0.8	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	0.8	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	0.8	0.5	0.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.9	19.9	15.5	24.9	10.2	7.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.4	20.2	10.7	22.4	7.6	7.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.6	21	15.9	24.4	9.5	7.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32	20	10.2	22.4	7.7	7.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.3	22.2	14.9	23.9	10	7.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.2	19.9	10.4	20.7	8.3	7.5
U2 - Efficacy additive alpha prior Beta(1, 1)	22.9	19.8	14.8	23.8	10.9	7.8
U3 - Efficacy additive alpha prior Beta(1, 1)	31.9	19.4	11.6	21.5	7.8	7.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.6	6.1	6	5.5	3.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.6	6	6.1	5.4	3.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.6	6	6.1	5.5	3.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.5	6.1	6.1	5.4	3.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.5	6.1	6.1	5.4	3.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.5	6	6.1	5.4	3.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.6	6	6	5.6	3.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.5	6	6	5.4	3.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.4	1.5	1.9	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.4	1.5	1.8	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.4	1.5	1.8	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.4	1.5	1.8	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.4	1.5	1.8	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.4	1.5	1.9	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.4	1.5	1.8	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.4	1.5	1.8	0.6	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	49.2	30.7	9.3	3.1	0.3	7.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	58.3	28	4.9	1.2	0.2	7.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	50	31.8	7.8	2.5	0.4	7.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	59.9	27.1	4.4	0.9	0.3	7.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	50.9	30.8	8	2.5	0.4	7.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	61.1	26.2	4	1.1	0.3	7.3
U2 - Efficacy additive alpha prior Beta(1, 1)	52.1	29.6	8.1	2.3	0.5	7.4
U3 - Efficacy additive alpha prior Beta(1, 1)	61.6	25.7	4.1	1.2	0.1	7.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.4	8	5.3	2.5	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.4	8	5.3	2.5	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.4	7.9	5.4	2.5	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.4	7.9	5.3	2.6	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.5	7.7	5.5	2.5	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.5	7.8	5.4	2.5	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.6	7.7	5.4	2.5	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.6	7.7	5.5	2.5	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.7	1.4	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.6	1.4	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.7	1.4	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.7	1.4	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.7	1.4	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.7	1.4	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.7	1.4	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.7	1.4	1.1	0.2	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.3	29.9	15.9	15.8	9	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.2	33.4	14.5	11.6	5.2	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.5	27.6	16.7	16.4	9.7	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	37	31.1	13.7	13.4	4.7	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.2	24.6	17.4	17.1	9.6	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	38.5	28.7	15.8	12.2	4.7	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	32.6	24.1	17.6	16.7	8.9	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	39.3	28	15	12.7	4.9	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.8	5.9	5.6	6	6.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.8	5.9	5.5	6	6.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	5.8	5.5	5.9	6.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	5.8	5.5	5.8	7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.9	5.6	5.6	5.9	7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.9	5.7	5.5	5.8	7	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.9	5.7	5.5	5.9	7	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.9	5.6	5.5	6	7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.2	1.8	2.7	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.3	1.8	2.7	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.3	1.8	2.6	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.3	1.8	2.6	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.2	1.8	2.6	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.2	1.8	2.6	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3.2	1.8	2.6	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	3.2	1.8	2.6	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.9	54	14.7	1.1	0	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	33.2	54.1	11.9	0.5	0	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.1	52.8	15.6	1.2	0	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.6	53.3	11.8	1	0	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.1	52.5	14.9	1.2	0	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.2	53.2	12.5	0.8	0	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	31	51.9	15.8	1	0	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	34.4	52.2	12.2	0.9	0	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	10.7	6.6	1.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	10.7	6.5	1.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11	10.6	6.5	1.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	10.5	6.6	1.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	10.3	6.7	1.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	10.3	6.6	1.6	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.2	10.3	6.7	1.6	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.2	10.2	6.7	1.6	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.9	1.1	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3	1.1	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1.1	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1.1	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1.1	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1.1	0.5	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	11.1	11	19.2	48.6	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.1	13.4	12.3	17.3	42.2	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	11.6	10.6	18.4	49.8	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.8	13.2	10.3	16.7	44.3	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.4	10	12.9	17.3	50.7	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.5	12.8	12.2	15.2	44.6	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	7.4	12.1	12	16.3	51.5	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	14	13.5	11.3	15.2	45.3	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	4.1	5	6	11.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	4	5	6	11.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	4	4.9	6	11.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	4	5	5.9	11.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	3.9	4.9	5.9	11.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	4	4.9	5.9	11.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.9	4	5	5.8	11.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.8	4	5	5.8	11.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.4	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.4	2.4	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.4	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.4	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.4	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.4	2.4	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.2	54.4	13.7	0.6	0	9.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.8	53.2	12.3	0.6	0	9.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.3	53.6	14.4	0.7	0	9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.5	52.9	11.9	0.7	0	9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.5	54.1	14.5	0.8	0	9.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.9	52.5	13	0.6	0	9
U2 - Efficacy additive alpha prior Beta(1, 1)	21.2	53.9	15.1	0.8	0	9
U3 - Efficacy additive alpha prior Beta(1, 1)	24.7	53.2	12.5	0.6	0	9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	10.9	6	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	10.9	6.1	1.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.4	10.9	6	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	10.8	6.1	1.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.4	10.7	6.1	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.4	10.7	6.1	1.4	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.4	10.7	6.2	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.5	10.6	6.2	1.4	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.7	1.2	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.7	1.2	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.7	1.1	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.7	1.2	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.7	1.1	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.7	1.1	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.7	1.2	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.7	1.1	0.5	0	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.6	49.4	25.7	6.5	0.5	4.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.5	51.5	21.1	6.2	0.4	4.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.4	48	26.6	7.1	0.6	4.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.1	49.9	22.1	5.5	0.1	4.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.8	47.6	27.6	7.2	0.4	4.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.3	48.8	22.3	5.1	0.2	4.3
U2 - Efficacy additive alpha prior Beta(1, 1)	11.9	48.2	28.3	6.7	0.6	4.3
U3 - Efficacy additive alpha prior Beta(1, 1)	19.6	48	22	5.8	0.3	4.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.2	9.7	8.1	3.8	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.1	9.8	8.2	3.8	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.2	9.7	8.3	3.8	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.2	9.6	8.4	3.7	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.2	9.5	8.4	3.8	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.2	9.5	8.4	3.8	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.2	9.5	8.4	3.8	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.2	9.3	8.5	3.9	1.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.8	1.7	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.8	1.7	1.4	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.8	1.7	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.8	1.7	1.4	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.8	1.7	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.8	1.7	1.4	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.8	1.7	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.8	1.7	1.4	0.3	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	58	1.6	0	0	0	40.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	58.1	1.6	0	0	0	40.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	58.6	1.4	0	0	0	40
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	58.7	1.4	0	0	0	39.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	58.5	1.2	0	0	0	40.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	58.6	1.3	0	0	0	40.1
U2 - Efficacy additive alpha prior Beta(1, 1)	58.6	1.2	0	0	0	40.2
U3 - Efficacy additive alpha prior Beta(1, 1)	59	1.2	0	0	0	39.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.4	1.7	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.4	1.6	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.4	1.6	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.4	1.6	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.4	1.6	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.4	1.6	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	21.4	1.6	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	21.4	1.6	0.3	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	0.6	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	0.6	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	0.7	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.5	58.8	10.6	0.5	0	4.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.5	57.8	8.8	0.4	0	4.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.1	59.5	11	0.7	0	4.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.9	55.3	8.7	0.6	0	4.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.1	58	11.6	0.7	0	4.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.2	56.2	8.3	0.6	0	4.7
U2 - Efficacy additive alpha prior Beta(1, 1)	24.5	59	11.1	0.7	0	4.7
U3 - Efficacy additive alpha prior Beta(1, 1)	30.1	55.6	9.1	0.5	0	4.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	12.9	5.3	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	12.9	5.3	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.8	12.9	5.4	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.9	12.8	5.5	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.9	12.8	5.4	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	12.7	5.4	1	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	10	12.7	5.4	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	10	12.7	5.5	1	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.1	1.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4	1.1	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.1	1.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1.1	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.1	1.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.1	1.1	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.1	1.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.1	1.1	0.4	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17	6	0	0	0	77
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	5.5	0	0	0	77.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.6	6.2	0	0	0	77.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.8	5.9	0	0	0	77.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.5	6.3	0	0	0	77.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.2	5.3	0	0	0	77.5
U2 - Efficacy additive alpha prior Beta(1, 1)	16.7	6.1	0	0	0	77.2
U3 - Efficacy additive alpha prior Beta(1, 1)	17	5.7	0	0	0	77.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.4	2.1	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.4	2.1	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.4	2.1	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.5	2.1	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.5	2.1	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.4	2.1	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.4	2.1	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.4	2.1	0.3	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.3	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.3	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.4	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.3	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.3	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.3	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.4	0.1	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	1.5	5.6	52.4	39.3	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	2	2.9	6.7	53.5	34.9	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	1.3	5	50.1	42.4	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	2.2	2.2	6.6	52	37	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	1.3	5.1	47.9	44.4	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.4	2.3	6.6	49.6	39.1	0
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	1.3	5.1	47.5	44.9	0
U3 - Efficacy additive alpha prior Beta(1, 1)	2.3	2.4	6.8	49.2	39.3	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.2	3.2	3.8	9	10.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.2	3.1	3.8	9	10.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.2	3.2	3.8	8.8	11.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.2	3.1	3.8	8.9	10.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.2	3.1	3.8	8.7	11.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.2	3.2	3.8	8.6	11.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.2	3.1	3.8	8.6	11.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.2	3.1	3.8	8.6	11.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.5	1.2	3.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.1	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.5	1.2	3.1	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.5	1.2	3.1	1.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.8	35.8	26.6	4.2	0	0.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	39.3	33.6	23	3.4	0	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.4	33.9	28.9	3.1	0	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	39	33.7	24	2.6	0	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.8	33.1	28.6	2.8	0	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	40.9	32	23.5	2.9	0	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	35.1	32.9	28.3	3.1	0	0.6
U3 - Efficacy additive alpha prior Beta(1, 1)	41.2	31.6	23.7	2.8	0	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	9.3	8.2	2	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	9.2	8.3	2	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.1	8.9	8.5	2.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.2	8.9	8.5	2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	8.8	8.5	2.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	8.8	8.4	2.1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.3	8.8	8.5	2.1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.3	8.8	8.5	2	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.3	1.4	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.3	1.5	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.2	1.4	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.2	1.4	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.2	1.4	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.2	1.4	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.3	1.4	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.3	1.4	0.7	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.1	32.4	21.8	3.8	0	10.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.2	33.3	18.7	2.9	0	10.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	31.2	32.3	22.2	3.5	0	10.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.6	33	19.7	2.9	0	10.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	32	32.4	21.6	3.2	0	10.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	35.9	31.1	19.2	2.9	0	10.9
U2 - Efficacy additive alpha prior Beta(1, 1)	31.8	31.8	22.5	3	0	10.9
U3 - Efficacy additive alpha prior Beta(1, 1)	36.2	30.1	19.6	3.2	0	10.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.2	7.7	6.7	1.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.2	7.8	6.6	1.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.3	7.6	6.7	1.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.3	7.6	6.7	1.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.4	7.4	6.8	1.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.4	7.4	6.7	1.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.4	7.4	6.7	1.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.4	7.4	6.8	1.9	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.1	1.1	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.1	1.1	0.7	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.1	1.1	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.1	1.1	0.7	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.1	1.1	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.1	1.1	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.1	1.1	0.7	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.1	1.1	0.7	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.8	28.8	37.3	6.4	0.2	5.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.3	30.1	33.9	5.1	0.1	5.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.1	28.3	39.4	5.5	0.2	5.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.3	28.1	35.7	5.3	0.1	5.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.7	27.9	39.4	5.3	0.2	5.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.6	26.8	37	5	0.1	5.5
U2 - Efficacy additive alpha prior Beta(1, 1)	21.3	27.1	40.9	5.1	0.1	5.5
U3 - Efficacy additive alpha prior Beta(1, 1)	26.7	27	36.3	4.4	0.1	5.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	7.4	9	3.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	7.5	9	3.2	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	7.4	9.1	3.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	7.4	9.2	3.1	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	7.3	9.1	3.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9	7.2	9.2	3	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.9	7.2	9.2	3.1	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	9	7.2	9.2	3.1	0.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.1	1.6	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.1	1.5	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.1	1.5	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.1	1.5	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.1	1.5	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.2	1.5	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.1	1.5	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.2	1.5	0.9	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.8	38.7	3	0	0	36.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.4	38.2	2.9	0	0	36.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.3	38.6	3.4	0	0	36.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.7	38.7	3	0	0	36.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.3	37.7	3.4	0	0	36.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.6	37.4	2.6	0	0	36.4
U2 - Efficacy additive alpha prior Beta(1, 1)	21.9	38	3.6	0	0	36.5
U3 - Efficacy additive alpha prior Beta(1, 1)	23.5	37.4	2.6	0	0	36.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.6	8.1	2.9	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.6	8.1	3	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.7	8	3	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.6	8	3	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.7	7.9	3	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.6	8	3	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	13.7	7.9	3	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	13.7	7.9	3	0.5	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.7	0.6	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.7	0.6	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.7	0.6	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.7	0.6	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	0.6	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	0.6	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	0.6	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	0.6	0.2	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	67.9	26.7	2.4	0	0	3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.8	24.9	2.2	0	0	3.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	69.7	24.8	2.5	0	0	3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	71.7	23	2.3	0	0	3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	70.6	24	2.5	0	0	2.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	73	22.4	1.5	0	0	3.1
U2 - Efficacy additive alpha prior Beta(1, 1)	70.8	24	2.3	0	0	2.9
U3 - Efficacy additive alpha prior Beta(1, 1)	73.7	21.9	1.4	0	0	3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.2	7.3	2.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19	7.4	2.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.3	7.2	2.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.3	7.1	2.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.5	7	2.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.5	7	2.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	19.4	7	2.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.4	7	2.7	0.3	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.1	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.1	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.1	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.1	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.1	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.1	0.6	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19	19.6	16.6	35.9	8.1	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.1	22.9	13.3	31.8	6.1	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.5	19	16.4	36.3	8	0.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.3	20.8	14.2	31	6.9	0.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.2	17.2	19.3	35.3	8.2	0.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.5	20.2	14.8	32.2	6.5	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	19.1	17.6	18.4	35.2	8.9	0.8
U3 - Efficacy additive alpha prior Beta(1, 1)	27.2	18.8	15.8	30.2	7.2	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	5.1	6.4	8.1	4.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	5.1	6.4	8.1	4.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	5	6.4	8.1	5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.4	5	6.4	8.2	4.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.4	5	6.4	8.1	5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.4	5	6.3	8.1	5	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.5	5	6.4	8	5	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.5	4.9	6.3	8.1	5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3	1.7	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.9	1.7	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1.6	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1.6	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.9	1.6	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1.6	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.9	1.7	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.9	1.6	2.3	0.9	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.6	35.9	27.6	24.7	6.1	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.4	42.2	25.2	19.9	4.2	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	35	29.1	24.4	5.5	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	40.4	27.1	20.3	3.3	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.2	33.2	30.8	23.8	5.9	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.1	38.4	29.2	19.8	3.4	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	6.5	32.1	30.4	24.9	6	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	8.6	38.8	28.7	20.4	3.4	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	5.9	6.7	8	5.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	5.9	6.6	8.1	5.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	5.9	6.7	8	5.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	5.8	6.7	8.1	5.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	5.8	6.8	7.9	5.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	5.8	6.7	8	5.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.1	5.8	6.8	7.9	5.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.1	5.8	6.8	8	5.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.4	1.8	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.4	1.8	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	16.5	33.5	30.3	9.5	0.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.5	21.4	32.7	25.4	5.4	0.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	16.1	34.3	29.9	10.3	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.2	20	33.2	25.1	5.9	0.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	14.7	35.8	27.3	11.9	0.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.6	19.7	35.1	22.5	6.5	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	15	36.8	26.6	11.4	0.6
U3 - Efficacy additive alpha prior Beta(1, 1)	15.8	18.7	35.6	23.3	6	0.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	4.5	7.9	7.7	6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	4.5	7.8	7.7	6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	4.4	7.9	7.7	6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	4.4	7.9	7.7	6.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	4.4	8	7.5	6.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	4.4	8	7.4	6.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.9	4.4	8	7.5	6.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.9	4.4	8	7.5	6.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2	2.7	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2	2.7	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	2	2.7	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	2	2.7	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	2	2.7	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	2	2.6	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	2	2.6	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	2	2.7	1.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7	8	15	44.5	25.5	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13	9.7	14.3	42.3	20.7	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.2	7.5	15.6	43.3	26.4	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12	9.4	14.5	42.2	21.9	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.6	8.2	15.1	43.9	26.2	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.6	8.7	14.7	42.2	21.8	0
U2 - Efficacy additive alpha prior Beta(1, 1)	6.2	8.8	14.8	44	26.2	0
U3 - Efficacy additive alpha prior Beta(1, 1)	12.7	9.4	13.4	42.6	21.9	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.7	5.5	8.6	8.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.7	5.5	8.6	8.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.7	5.4	8.5	8.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.6	5.6	8.4	8.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.7	5.4	8.4	9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.6	5.5	8.5	8.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.6	5.4	8.5	8.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.7	5.5	8.4	8.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.9	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.9	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.8	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.8	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.8	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.8	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	2.8	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	2.8	0.9	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.8	16.6	49	13.2	3.4	2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.3	18.4	43.8	12.2	2.3	2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.2	16.8	48.5	14	3.5	2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22	18.2	43.5	11.9	2.4	2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.3	16.5	49	13.6	3.6	2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.1	17	42.9	12.1	2.9	2
U2 - Efficacy additive alpha prior Beta(1, 1)	14.6	18	47.6	14.6	3.2	2
U3 - Efficacy additive alpha prior Beta(1, 1)	23.6	16.8	42.8	11.5	3.3	2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.7	6.7	10.1	5	2.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.7	6.6	10.2	5	2.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.8	6.6	10	5	2.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.8	6.6	10.1	5	2.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	6.4	10.3	4.9	2.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	6.6	10.2	4.9	2.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.7	6.5	10.2	5	2.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.7	6.5	10.2	4.9	2.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.4	2	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.4	2	1.5	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.4	1.9	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.4	1.9	1.5	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.4	1.9	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.4	2	1.5	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.4	2	1.5	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.4	2	1.5	0.5	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	45.1	31.2	10.6	1.6	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.8	49.2	26.3	8.3	1.3	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	43.3	32.7	10.3	2.2	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.7	46.1	27	8.8	1.3	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.4	41	33.7	10.6	2.2	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.9	44.9	27.4	8.9	1.8	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	12.5	42	32.6	10.3	2.5	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	16.8	46.2	27.1	8.2	1.6	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	9.1	9	4.9	2.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	9.1	9.1	4.9	2.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	8.9	9.2	5	2.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	8.9	9.2	4.9	2.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	8.7	9.4	4.9	2.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	8.8	9.3	4.8	2.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.7	8.7	9.3	4.9	2.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.7	8.7	9.4	4.9	2.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.2	1.6	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.2	1.6	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.2	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.2	1.6	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.2	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.2	1.6	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.2	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.2	1.6	1.3	0.4	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.1	46.8	11.2	2.8	0.9	16.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.8	45.3	9.4	2.8	0.5	16.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.8	45.7	11.8	2.7	0.9	16.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.9	44.5	9.2	2.5	0.7	16.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.2	46.6	11.3	2.9	0.9	16.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.4	45.4	9.1	2.2	0.7	16.2
U2 - Efficacy additive alpha prior Beta(1, 1)	22.4	46.6	11.3	2.7	0.9	16.1
U3 - Efficacy additive alpha prior Beta(1, 1)	27.8	44.1	9.1	2.2	0.8	16
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	8.9	5	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	8.8	5	2	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	8.8	5	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11	8.9	5	2	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	8.7	5.1	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	8.9	5	1.9	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.1	8.7	5	2	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.1	8.8	5	2	0.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.6	1.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.7	1	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.7	1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.7	1	0.7	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.6	56.6	9.3	1.3	0.1	7.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.3	52.6	7	0.8	0.1	7.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.5	54.9	10.2	1.1	0.1	7.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.3	51.2	8.2	0.9	0.1	7.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.8	54.6	10.4	1.1	0	7.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.4	53.3	7.5	0.6	0	7.2
U2 - Efficacy additive alpha prior Beta(1, 1)	25.5	55.7	10.3	1.4	0	7.1
U3 - Efficacy additive alpha prior Beta(1, 1)	32.4	51.7	7.7	1	0	7.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	11	5.6	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	11	5.7	1.6	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.8	11.1	5.6	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.9	11	5.6	1.6	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.9	10.9	5.7	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.9	11	5.6	1.6	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.9	10.8	5.7	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.9	10.9	5.6	1.6	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.1	1.2	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.1	1.2	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.1	1.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.2	1.1	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.1	1.2	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.2	1.2	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.2	1.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.2	1.1	0.7	0.1	